

A mesh-free redistribution method for the simulation of diffusion in arbitrary domains

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This paper describes a mesh-free redistribution method for the simulation of diffusion in arbitrary bounded domains which is compatible with Lagrangian vortex methods. Lagrangian mesh-free methods based on the vorticity formulation of the Navier Stokes equations have been particularly attractive in the simulation of high Reynolds number flows that experience strong dynamic changes which are challenging to capture using grid-based methods with static meshes. The simulation of diffusion in these mesh free methods remains a challenge especially in bounded domains with complex boundaries. The redistribution method in [1] assumes that the flow domain is locally semi-infinite and is bounded by the tangent to the solid boundary at the point nearest to the element. The error involved in diffusing an element of cutoff radius σ at a distance y_0 from a solid boundary of local radius of curvature R grows with $(R/\sigma)^{-1}$ and $(y_0/\sigma)^{-1}$. To keep the error small, elements near the boundary must be assigned small σ , which in view of the overlap requirement, results in smaller elements spacing near the boundary and leads to a substantial growth in computational cost.

The generalized redistribution scheme proposed here does not make any assumptions regarding the solid boundary geometry. The equations governing the fractions of the element's strength distributed to its neighbors during diffusion are obtained by computing the moments of the diffusion equation, $\frac{\partial}{\partial \tau} \int_{\Omega} x^m y^n \omega d\mathbf{x} = \int_{\Omega} x^m y^n \nabla^2 \omega d\mathbf{x}$. The integrand on the R.H.S. is replaced by $\nabla \cdot (x^m y^n \nabla \omega - \omega \nabla x^m y^n) + \omega \nabla^2 (x^m y^n)$. This leads to expressing the time variation of the (m, n) moment of vorticity in terms of a line integral over the boundary and the $(m-2, n)$ and $(m, n-2)$ moments,

$$\frac{dM_{m,n}}{d\tau} = \int_{\partial\Omega} (x^m y^n \nabla \omega - \omega \nabla x^m y^n) \cdot \hat{\mathbf{n}} dS + m(m-1)M_{m-2,n} + n(n-1)M_{m,n-2}$$

where $\tau = \nu t$. While these equations conserve the moments in the flow domain, they provide no answer for what happens to the vorticity that crosses the boundary for a given choice of the cutoff functions. For a zero flux boundary condition, this vorticity must be reflected back into the flow domain in a physically meaningful and a computationally accurate manner.

Accuracy and cost of the proposed scheme are compared to previous redistribution schemes [1] for the problem of diffusion of temperature in bounded domains of different geometries subject to zero heat flux at the solid boundary.

References

[1] I. Lakkis and A. Ghoniem, A high resolution spatially adaptive vortex method for separating flows. Part I: Two-dimensional domains, *J. Comput. Phys.*, 228(2):491–515, 2009.